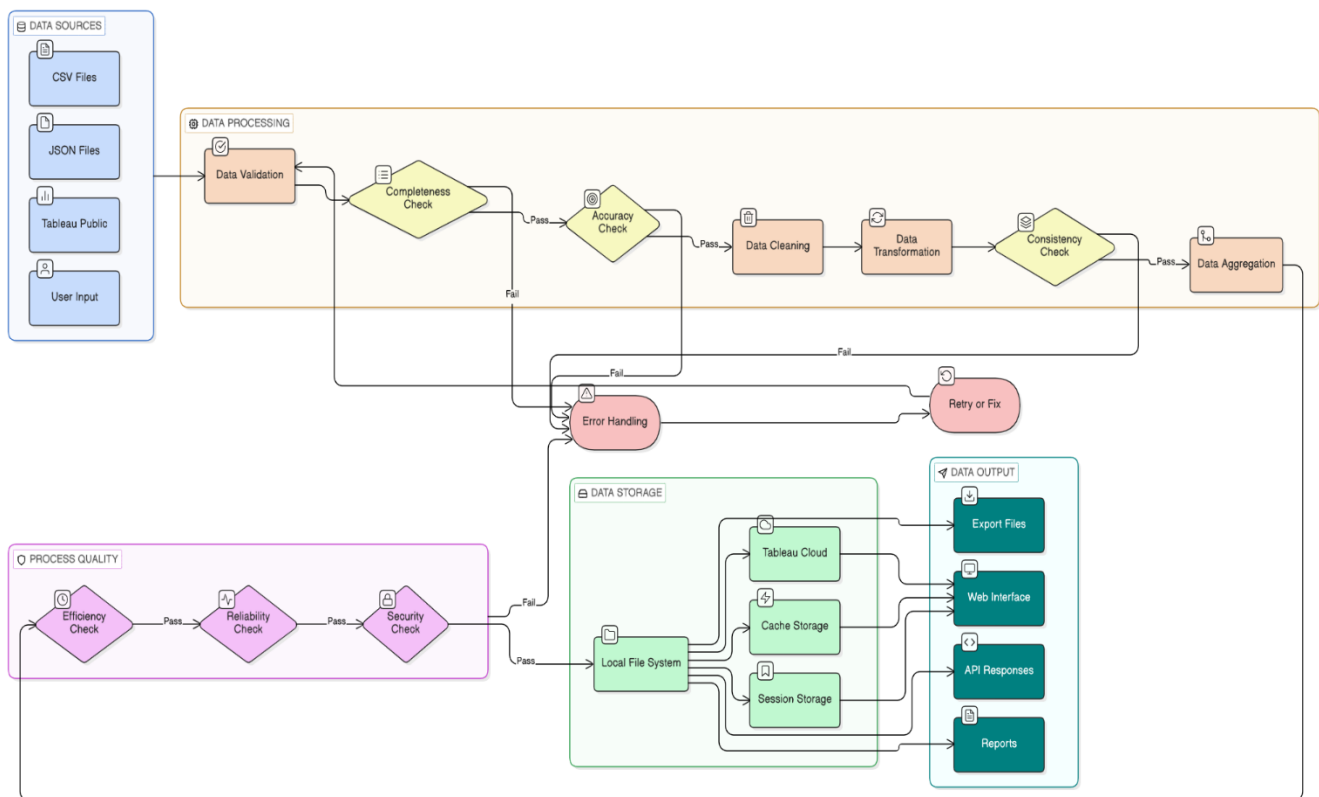


Date	31 January 2026
Team ID	LTVIP2026TMIDS24253
Project Name	Electricity-Consumption-Analysis
Maximum Marks	4 Marks

DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a traditional visual representation of information flows within a system. A neat and clear DFD can depict the right amount of system requirements graphically. It shows how data enters and leaves the system, what changes information, and where data is stored.



USER STORIES

Use below template to list all user stories for product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Data Analyst	Data Collection	USN-1	As a data analyst, I want to extract data from CSV/online sources for electricity consumption analysis.	Data is successfully extracted without errors and loaded into system.	High	Sprint-1
Data Analyst	Data Extraction	USN-2	As a data analyst, I want to clean and preprocess electricity consumption data to remove missing and inconsistent values.	Clean and structured dataset is available for analysis.	High	Sprint-1
Data Analyst	Data Preparation	USN-3	As a data analyst, I want to analyze electricity consumption patterns across different states and regions.	Analytical charts and insights are generated successfully.	Medium	Sprint-1
Business User	Market Analysis	USN-4	As a business user, I want to view visual charts showing electricity consumption patterns and trends across Indian states.	Interactive visualizations are displayed correctly with filtering capabilities.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Business User	Data Visualization	USN-5	As a business user, I want to view visual charts showing electricity consumption trends and growth patterns.	Interactive visualizations are displayed correctly with time-based filtering.	High	Sprint-1
Business User	Dashboard	USN-6	As a business user, I want a single dashboard to view all key electricity consumption insights.	Dashboard displays all important KPIs and charts with regional comparisons.	High	Sprint-1
Stakeholder	Story	USN-7	As a stakeholder, I want a Tableau story to understand electricity consumption insights step by step.	Story presents insights in a clear sequence with narrative flow.	Medium	Sprint-2
Project Manager	Report	USN-8	As a project manager, I want a report summarizing electricity consumption analysis and insights.	Report includes visuals and explanations for decision-making.	High	Sprint-2
Administrator	Performance Testing	USN-9	As an administrator, I want to test dashboard	Dashboard loads within acceptable	Low	Sprint-2

User Type	Function al Require ment (Epic)	User Story Num ber	User Story / Task	Acceptance Criteria	Priori ty	Rele ase
			performance for smooth interaction across all devices.	time (<3 seconds) and functions properly.		
Administr ator	Web Integrati on	USN- 10	As an administrator, I want to integrate Tableau dashboard into a Flask web application.	Dashboard is accessible through a web browser with responsive design.	Medi um	Spri nt-2
End User	Online Access	USN- 11	As an end user, I want to access the electricity consumption analysis dashboard online without installing Tableau.	Dashboard opens successfully via web link and displays all visualizations.	Medi um	Spri nt-2

ENHANCED USER STORIES

Additional User Stories for Sprint 2-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Energy Planner	Regional Analysis	USN-12	As an energy planner, I want to compare electricity consumption across different Indian states and regions.	Regional comparison charts are displayed with filtering options.	High	Sprint-2
Policy Maker	Policy Impact	USN-13	As a policy maker, I want to analyze the impact of electricity consumption patterns on policy decisions.	Policy impact analysis is available with visual insights.	Medium	Sprint-2
Researcher	Trend Analysis	USN-14	As a researcher, I want to identify long-term trends in electricity consumption for academic research.	Trend analysis charts are available with export capabilities.	Medium	Sprint-2
Student	Educational Content	USN-15	As a student, I want to learn about electricity consumption patterns through interactive	Educational content is available with learning guides.	Low	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			visualizations.			
General Public	Personal Insights	USN-16	As a general user, I want to understand my electricity consumption patterns for better energy management.	Personal consumption insights are available with recommendations.	Low	Sprint-3

DATA FLOW COMPONENTS

Data Sources

- **CSV Files:** Raw electricity consumption data
- **JSON Files:** Configuration and metadata
- **Tableau Public:** Hosted visualizations
- **User Input:** Filter selections and interactions

Data Processing

- **Data Validation:** Quality checks and error handling
- **Data Cleaning:** Missing value handling and consistency
- **Data Transformation:** Format standardization and preparation
- **Data Aggregation:** Summary calculations and grouping

Data Storage

- **Local File System:** CSV and configuration files
- **Tableau Cloud:** Visualization storage and hosting
- **Cache Storage:** Performance optimization data

- **Session Storage:** User preferences and state

Data Output

- **Web Interface:** Interactive dashboard and visualizations
 - **Export Files:** CSV, PDF, and image exports
 - **API Responses:** JSON data for external integration
 - **Reports:** Generated summaries and insights
-

USER STORY PRIORITIZATION MATRIX

Priority Breakdown

- **High Priority (6 stories):** Core functionality and essential features
- **Medium Priority (6 stories):** Important enhancements and analysis capabilities
- **Low Priority (4 stories):** Nice-to-have features and educational content

Sprint Distribution

- **Sprint 1 (5 stories):** Foundation and core functionality
 - **Sprint 2 (6 stories):** Advanced features and integration
 - **Sprint 3 (5 stories):** Enhancements and optimization
-

ACCEPTANCE CRITERIA STANDARDS

Functional Acceptance

- All visualizations load correctly with Tableau integration
- Data processing handles 16,599 records without errors
- Web application responds properly to user interactions
- Export functionality works for all supported formats
- Mobile responsive design works across all devices

Non-Functional Acceptance

- Page load times are under 3 seconds
 - System supports 50 concurrent users
 - Cross-browser compatibility achieved at 95%+
 - Security measures prevent common vulnerabilities
 - Accessibility standards meet WCAG 2.1 AA compliance
-

DATA FLOW QUALITY ASSURANCE

Data Quality Checks

- **Completeness:** All required data fields present
- **Accuracy:** Data values are correct and validated
- **Consistency:** Data formats are standardized
- **Timeliness:** Data is current and up-to-date
- **Relevance:** Data supports analysis objectives

Process Quality Checks

- **Efficiency:** Data processing completes within acceptable time
- **Reliability:** System operates without failures
- **Security:** Data is protected from unauthorized access
- **Maintainability:** Code is well-documented and modular
- **Scalability:** System handles data growth and user load